

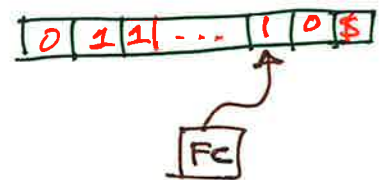
Finite Automata

This is our first and simplest computational model. A Finite Automaton (FA) M , ~~accepts~~ ~~recognizes~~ a language, usually written as $L(M)$, over an alphabet Σ .

That is, for each $x \in \Sigma^*$, M needs to decide whether to accept x or reject x .

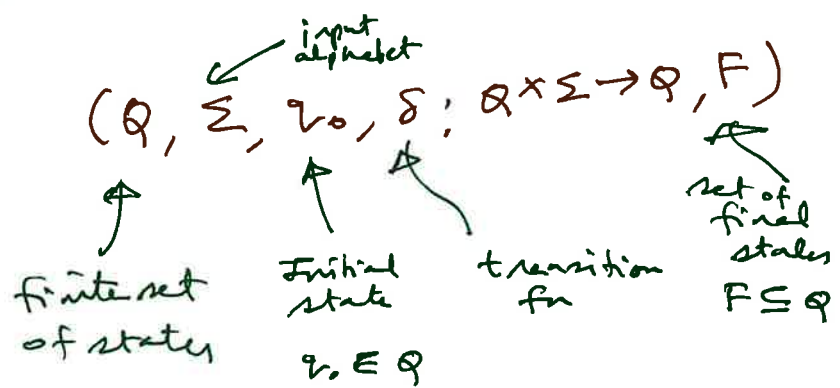
An FA makes this decision by a simple computation process described now.

Input is presented on a linear tape, with one symbol of Σ written in any cell, except the last cell which a special symbol $\$ \notin \Sigma$ written on it.



FA M has a finite state control with one read only head attached to it. The head scans a cell on the input tape, in any configuration.

M is specified as a five tuple



Initially, M is in state q_0 and ^{input} head is positioned on the leftmost symbol. In general, when the head is reading 'a' and M is in state 'q' then M ~~moves to~~ changes its state to $\delta(q, a)$ and ^{its head} moves one ^{cell} to the right on the input tape.

When M 's head scans '\$' (the endmarker) then it stops (without changing the ~~state~~ state). The ^{input} is accepted iff state of $M \in F$ when it halts.

Example 1:

Let us design an automaton M , which accepts a string $x \in \{0,1\}^*$ iff it has an even no. of

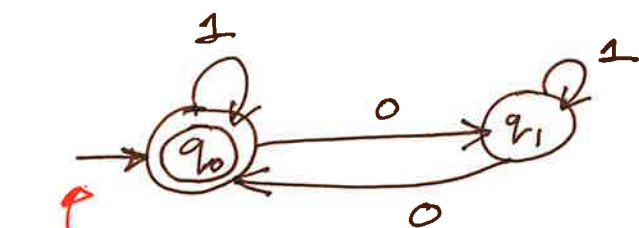
0's. We take $Q = \{q_0, q_1\}$, $F = \{q_0\}$

$$\delta(q_0, 1) = q_0 \quad \delta(q_1, 1) = q_1$$

$$\delta(q_0, 0) = q_1 \quad \delta(q_1, 0) = q_0$$

our states simply record parity of the no. of 0's seen in the input so far.

There is a more pictorial way to present automaton M , as a transition diagram.



↑
represents
start state

Double circle
represents a final
state.

Arrows are transition
 $\delta(q, a) = q'$

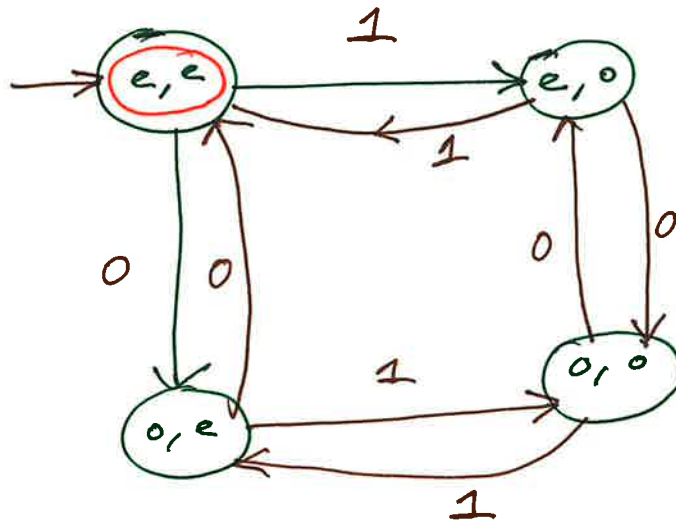
is represented as an
arrow from q to q'
labelled ' a '.

Example 2

Design an automaton M s.t.

$$L(M) = \{ w \in \{0,1\}^* \mid w \text{ has even no. of } 0\text{'s} \text{ and an even no. of } 1\text{'s} \}$$

Solution



Meaning of states

(0, 2)

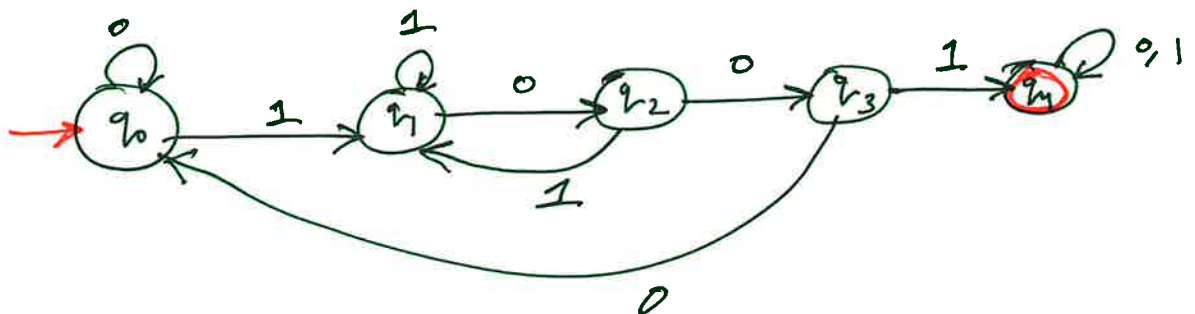
odd ~~even~~ no. of
0's and an
even no. of 1's
[0: odd
e: even]

Example 3

Design an automaton M s.t.

$$L(M) = \{ w \in \{0,1\}^* \mid w \text{ has a substring } 1001 \}$$

Solution

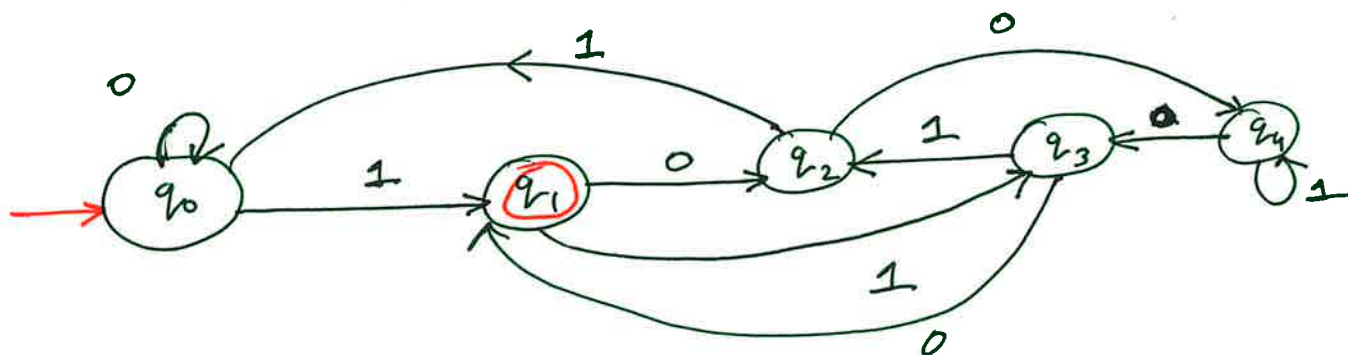


Example 4

Design an automaton M s.t.

$$L(M) = \{ w \in \{0,1\}^* \mid w \bmod 5 = 1 \}$$

when w is considered as a binary string with MSB as the leftmost bit



Observation:

if w is n then $n0 = 2n$

$$n1 = 2n + 1$$

$$'n0' \bmod 5 = 2n \bmod 5 = [2(n \bmod 5)] \bmod 5$$

$$'n1' \bmod 5 = (2n + 1) \bmod 5 = [2(n \bmod 5) + 1] \bmod 5$$

We can also extend $\delta: Q \times \Sigma \rightarrow Q$ to

$\hat{\delta}: Q \times \Sigma^* \rightarrow Q$ as follows

$$\hat{\delta}(q, \epsilon) = q$$

$$\hat{\delta}(q, xa) = \delta(\hat{\delta}(q, x), a) \quad [x \in \Sigma^*, a \in \Sigma]$$

$\hat{\delta}(q, x)$ represents the state of automaton after reading ~~x~~ if the initial state was q .

$$\begin{aligned} \hat{\delta}(q, a) &= \delta(\hat{\delta}(q, \epsilon), a) && \text{by defn of } \hat{\delta} \text{ as } a = \epsilon a \\ &= \delta(q, a) && \text{by defn of } \delta \end{aligned}$$

$\Rightarrow \hat{\delta}$ and δ ~~coincide~~ coincide on common part, $Q \times \Sigma$.

We therefore use only one symbol, δ , for both fns.

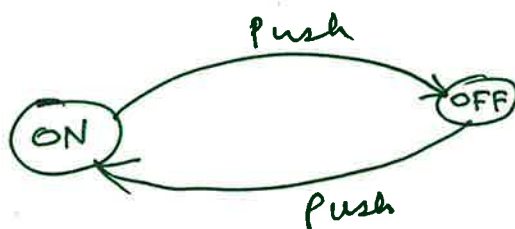
The language associated with the automaton is

$$\{x \in \Sigma^* \mid \delta(q_0, x) \in F\}$$

We have motivated and looked at FA so far as a recognizer of a language. However, FA is also used to represent (model/specify) finite state systems, with discrete ~~input~~ time inputs.

Example 1:

★ Consider a simple ON/OFF switch. It can be modeled as a two state FA



Example 2:

A Swing-door, with sensors on both (front and rear side). It's ~~has~~ sensors detect if someone is standing in front/rear side of the ~~door~~ entrance.

The input therefore consists of

Front (person only in front)

Rear

Both

Neither

(people both in front and ~~both~~ rear)

